Briana N.C. Chronister

EDUCATION

UNIVERSITY OF CALIFORNIA SAN DIEGO (UCSD)– SAN DIEGO STATE UNIVERSITY (SDSU) San Diego, CA

PhD in Public Health, Global Health Concentration August 2019 – September 2024

Dissertation Title: Leveraging Remote Sensing and Machine Learning to Assess the Impact of Home Proximity to Agriculture on Pesticide Exposure, Depression, and Anxiety in an Ecuadorian Agricultural Community

Dissertation Committee: Dr. Jose Ricardo Suarez-Lopez (Chair; UCSD), Dr. Raeanne C. Moore (UCSD), Dr. Tommi Gaines (UCSD), Dr. Humberto Parada (SDSU), Dr. Penelope (Jenny) Quintana (SDSU), Dr. Dan Sousa (SDSU)

T32 TADA-BSSR Pre-doctoral Trainee (5T32MH122376)

UNIVERSITY OF NOTRE DAME Notre Dame, IN

Master of Science in Global Health July 2015

Advisor: Dr. Shaun Lee

Thesis: Implementation of an antibiotic education program for community health promoters and an assessment of antibiotic knowledge in Tena and Quito, Ecuador

Bachelor of Science in Biological Sciences May 2014

Minor: Science Technology and Values

RESEARCH EXPERIENCE

University of California San Diego

Postdoctoral Researcher – Mentors: Dr. Raeanne C. Moore and Dr. Jose Suarez Oct 2024 – Present

Study Title: Cyclical alterations in neurocognitive performance and mood over two years in relation to pesticide spray seasons among adolescent and young adult participants of the ESPINA study (EPSPINA-EMCT). La Jolla, CA

- Plans, conducts, and evaluates epidemiologic investigations on cyclical alterations in neurocognitive performance and mental health over a two-year period in adolescent and young adult participants of the ESPINA-EMCT study, with emphasis on population-based surveillance and prevention of health disparities related to pesticide exposure.
- Examines environmental and occupational determinants of pesticide exposure and their impacts across the life course, aligning with health equity goals by investigating vulnerable populations in agricultural communities.
- Monitors, analyzes, and reports trends related to pesticide spray seasons, leveraging longitudinal surveillance data to evaluate potential neurobehavioral and mental health outcomes of environmental exposures.
- Applies advanced statistical methods (R, SAS, Python) and interdisciplinary approaches to address complex public health challenges, consistent with the County's focus on data-driven decision-making and performance accountability.
- Designs and executes statistical analysis plans and develops epidemiologic reports, including analysis-ready datasets, tables, and visualizations, supporting transparency, evaluation, and continuous quality improvement.
- Conducts systematic literature reviews to strengthen evidence-based practice, inform study design, and contextualize findings; disseminates results through peer-reviewed publications, policy-relevant reports, and presentations at national and international conferences.
- Supervises and mentors trainees, strengthening the public health workforce and fostering leadership skills among undergraduate and graduate research assistants.
- Examines associations with pesticide spray seasons, leveraging longitudinal data to understand potential neurobehavioral effects of environmental exposures.
- Applies advanced statistical methods and interdisciplinary approaches to address environmental health challenges within neuroscience topic areas.
- Designed and executed statistical analysis plans and produced statistical programming deliverables, including cleaned, analysis-ready datasets, tables, and visualizations.
- Conducts comprehensive literature reviews to inform study design and contextualize findings; develops and publishes scientific manuscripts in peer-reviewed journals and presents research through posters and oral presentations at national and international conferences.

Graduate Student Researcher – Principal Investigator: Dr. Jose Suarez Aug 2019 – Sep 2024

Study Title: Secondary Exposure to Pesticides among Children and Adolescents (ESPINA) La Jolla, CA

- Conducted cross-sectional (e.g., linear, and logistic regression, principal component analysis, Bayesian Kernel Machine Regression) and longitudinal (e.g. Generalized Estimating Equations, Kaplan-Meier tests, cox-regression) data analysis of the ESPINA cohort to understand the associations of sub-clinical exposures to pesticides with short- and long-term alterations in the development of children living the agricultural county of Pedro Moncayo, Ecuador. Analysis is conducted in SAS and R.
- Created visualizations in SAS and R to appropriately display analysis results; this includes graphing locally estimated scatterplot smoothing graphs, creating boxplots, and scatter plots.
- Conducted spatial analysis (e.g., geospatially weighted regression, cluster analysis, hotspot analysis) to determine the influences of the built environment on pesticide exposure and health outcomes using R, ArcGIS, QGIS, and Python.
- Conducted image processing on Sentinel-2 and Planet satellite images, by applying Acolite processing and spectral unmixing.
- Leveraged machine learning and artificial intelligence computer vision methods, like Uniform Manifold Approximation and Projection for Dimension Reduction (UMAP) and object segmentation using the Meta AI Segment Anything Model, on Sentinel-2 and Planet images using Python and Linux to identify where agriculture is present in Pedro Moncayo, Ecuador.
- Applied environmental and spatial epidemiology methods to determine whether there are associations between pesticide exposure with adrenal and gonadal hormone concentrations, depression (CDI-2) and anxiety (MASC-2) symptoms, and neurobehavioral changes (measured using the NEPSY-II).
- Conducted fieldwork activities in Pedro Moncayo and Quito, Ecuador to meet project goals, to administer neurobehavioral exams in Spanish such as the NIH Toolbox and NeuroUX ecological momentary assessment tests, measure acetylcholinesterase activity, and collect ground truth land cover data.
- Prepared and submitted an Institutional Review Board (IRB) modification application for my created study protocol that implemented location tracking measures for assessment periods. This included amending the protocol, creating new data collection instruments, drafting consent form documents, translating documents to Spanish, and submitting the amendment within the Kuali platform.
- Built a REDCap database that captured participant data and e-consents using appropriate e-consenting frameworks.

Graduate Student Researcher – Principal Investigator: Dr. Christine Thorne December 2023 – Sep 2024
Study Title: Tele-ophthalmology for Detection of Diabetic Retinopathy La Jolla, CA

- Designed and implemented a data analysis plan to understand patterns of access to and follow-up from teleophthalmology sites to improve eye screening and reduce morbidity associated with vision loss for diabetic patients that are prone to diabetic retinopathy.
- Implemented spatial analysis techniques, such as network analysis, to determine whether participant home distance to teleophthalmology sites is associated with retention to care and vision loss for diabetic patients.
- Conducted data management and data cleaning for project databases.
- Coordinated with the Collaborative UC Tele-ophthalmology Initiative (CUTI) sites, such as the ophthalmology team at the University of California Davis, to augment and align analysis goals.

Graduate Student Researcher – Principal Investigator: Dr. Sheri Hartman June 2020 – October 2020
Study Title: Improve Cognition after Breast Cancer (I Can!) La Jolla, CA

- Assisted with recruitment, enrollment, and retention of 200 study participants by using the California Cancer Registry, a research study in partnership with the Oncology Department (Moore's Cancer Center) and the UC San Diego Public Health department.
- Created standard operating procedures for assessment activities used as a reference to train new staff.
- Conducted data cleaning and validation of the REDCap databases.
- As the lead measurement role, administered cognitive assessments to assess participant's progress. These assessments include the NIH Toolbox, Conners Continuous Performance Test (CPT-3), National Adult Reading Test (NART), and

Hopkins Verbal Learning Test (HVLT).

University of Notre Dame

Graduate Researcher – Principal Investigator: Dr. Shaun Lee University of Notre Dame
Study Title: Antibiotic Resistance in Ecuador Aug 2014-July 2015

- Researched factors contributing to antibiotic self-medication and its impact on antibiotic resistance in Ecuador.
- Created a survey containing modules for demographics, behavior of antibiotic use, opinion of antibiotic resistance, and general knowledge of antibiotics.
- Collected survey data on populations in both Quito and Tena, Ecuador. Assessed regional differences and similarities of knowledge and use of antibiotics between urban and Amazon populations using SPSS.
- Identified education level and city location as significant factors in predicting an individual's antibiotic knowledge, with a positive correlation between appropriate use of antibiotics and knowledge.
- Drafted, implemented, and evaluated an antibiotic education program for Community Health Workers working with Timmy Global Health in Quito, Ecuador over a seven-week period.
- Wrote Master's Capstone thesis on the findings of the research project.

SELECT WORK EXPERIENCE

University of California San Diego Jan 2022 – Mar 2022

Instructional Teaching Assistant, San Diego, CA

- FMPH 404–Introduction to Environmental and Occupational Health (Winter Quarter)
  - Lead lab discussions, graded assignments, provided students with in-class support, answered student questions, and held weekly office hours.

San Diego State University Aug 2020 – May 2022

Graduate Teaching Assistant San Diego, CA

- PH 664 – Health, Society and Human Behavior (Spring Semester)
  - Assists project teams develop and implement a public health project in a San Quintin, Mexico Community. This includes providing expertise and feedback on project proposals and development.
- PH 620 – SAS for Biostatistics I (Fall Semester)
- PH 629 – SAS for Biostatistics II (Spring Semester)
  - Provided multiple guest lectures on macros, facilitated lab sessions, answered questions, graded assignments, provided students with support and feedback on assignments.

University of Virginia Nov 2017 – Aug 2019

Clinical Research Coordinator, Intermediate Charlottesville, VA

- Managed multiple clinical trials and device trials for investigators in the Division of Maternal-Fetal Medicine, within the Obstetrics and Gynecology Department.
- Created, managed, and analyzed real-world data sources (e.g. Electronic Health Records [EHR] and insurance claims) using STATA for both retrospective and prospective research studies.
- Recruited, screened, consented, and enrolled study participants who meet associated eligibility criteria for all studies.
- Maintained appropriate screening and enrollment logs, track participants, and prepare study materials.
- Prepared and submitted continuing reviews, modification submissions, reportable information, and Adverse Event submissions to the University of Virginia Institutional Review Boards.
- Assisted Principal Investigators in developing and reviewing clinical trial protocols to determine study feasibility.
- Submitted all contracts and material transfer forms to the UVa Office of Grants and Contracts for approval.
- Prepared and negotiated study budgets for all projects.
- Properly maintained regulatory information and patient study files in compliance with University and Federal policies of Patient Health Information Protection.
- Coordinated and attended site evaluation and initiation meetings.
- Managed the Maternal-Fetal Medicine Biological Specimen laboratory and maintained responsibility for patient identification, and specimen verification. Maintained all laboratory equipment, including but not limited to centrifuges,

pipettes, and deep freezers.
- Trained laboratory personnel and healthcare providers in collection and preservation techniques for specimens.
- Conducted specimen collection of biological samples according to study protocol.
- Processed, prepared, delivered, and/or shipped biospecimen samples to collaborators as required based on study protocol.

Icahn School of Medicine at Mount Sinai Apr 2017 – Nov 2017
Program Manager I New York, NY
- Led cross-functional research teams of up to 20 staff members to successfully execute multiple NIH-, NHLBI-, and AHA-funded clinical research projects focused on chronic disease prevention and management (e.g., hypertension, diabetes).
- Oversaw full-cycle clinical project management including budget development, regulatory compliance, proposal submission, and stakeholder communication in alignment with institutional and sponsor requirements.
- Acted as the primary liaison between internal investigators, grant managers, and external collaborating institutions, ensuring the alignment of clinical project strategies and regulatory standards (e.g., NIH, IRB guidelines).
- Monitored study budgets and expenditures, ensuring accurate charging of activities, proactive discrepancy reporting, and financial reconciliation against sponsor and institutional ledgers.
- Facilitated interim and final project reviews, analyzed and interpreted study data, and prepared data for regulatory reporting and publication.
- Provided leadership in issue resolution during project periods, including identification of risks, contingency planning, and proactive communication with executive-level stakeholders.
- Managed and ensured compliance of all subcontractor documentation and award agreement terms and conditions prior to study execution.
- Supported continuous process improvement initiatives by streamlining research administration processes and enhancing compliance oversight.
- Projects included National Institute of Health (NIH), National Heart, Lung, and Blood Institute (NHLBI), and American Heart Association (AHA) funded studies aiming to improve chronic conditions (i.e., Hypertension and Diabetes)(prevention, care and/or management. These project Optimizing Linkage and Retention for Hypertension Care in Rural Kenya (LARK), Bridging Income Generation with GrouP Integrated Care (BIGPIC), Nurse Management of Hypertension in Rural Western Kenya, and Family-Based Approach in a Minority Community Integrating Systems-Biology for Promotion of Health (FAMILIA).

Program Coordinator I Nov 2015-Apr 2017
- Coordinated clinical research activities across multiple departments, ensuring program objectives were met efficiently and in compliance with regulatory standards.
- Assisted in the development of research protocols, IRB submissions, and grant applications, contributing to the design and execution of clinical studies.
- Led operational support for data collection procedures, statistical data analysis (quantitative and qualitative), and the preparation of figures and manuscripts for scientific dissemination.
- Supported grant reporting, project status monitoring, and helped identify risks and solutions to maintain study timelines and deliverables.
- Collaborated with internal and external stakeholders to optimize study operations and contributed to the continuous improvement of research management processes.
- Supported projects addressing chronic disease interventions through interdisciplinary and community-based research programs in international settings.

The Tempositions Group of Companies: School Professionals Nov 2016-Nov 2017
Assessor II New York, NY
- Coordinated and oversaw assessment events for research projects in New York.
- Assisted in project promotion in order to increase recruitment of families whose children attend in participating Head start schools in Harlem, New York.
- Collaborated with research staff to help create and design assessment tools such as baseline and follow-up

questionnaires to assess individual health behaviors and knowledge.
- Explained research activities in order to obtain informed consent of potential project participants.
- Obtained blood specimens from adult participants in order to compare epigenetic expression between caregivers and their children pre- and post-interventions.
- Assisted in conducting assessments by collecting measurement data such blood pressures, height, weight, blood samples and other routine measurements.
- Assisted in data entry validation and development of the project REDCap database.

LEADERSHIP & SERVICE

Joint Doctoral Program in Public Health Student Association 2020 – 2022

Racism in Public Health Seminar Series Organizer, October 2020 - 2022
- Organized the inaugural JDP in Public Health Student Association Quarterly Seminar Series. The theme for the inaugural seminar series was "Racism in Public Health."

ND Interconnected Program 2020-2021

Mentor, July 2020 – May 2021
- Provides mentorship to two first generation University of Notre Dame undergraduate students.

Timmy Global Health Club 2011 – 2014

Treasurer, Feb 2013-May 2014
- Participated in two weeklong medical brigades to Quito, Ecuador in the Fall of 2012 and 2013 with the goal of expanding health care to underserved populations.
- Translated during the doctor's consultations, obtained medical histories, and conducted lab work in Triage.
- Worked with club officers to organize PIH speaker Samson Njolomole's visit, volunteer opportunities, the annual brigade, fundraising events, and managed the club budget of more than $25,000.

MariachiND 2010 – 2014

Director, Feb 2013-May 2014; *President* Feb 2013-May 2014

Coro Primavera de Nuestra Señora, 2010-Spring 2014
- Delegated tasks and responsibilities to the club officers in order to increase efficiency and effective management.
- Increased the club's visibility by increasing both performances and social media presence, doubling the club size.
- Collaborated to write the club constitution and budget that elevated club status from probationary to full time.

RESEARCH PUBLICATIONS AND PRESENTATIONS

Publications

1. Pfaff M, Zambrano-Romero A, LeFebre GH, Ochoa-Herrera V, Gupta S, **Chronister BNC**, Mianecki AL, Carpintero-Salvador N, Checkoway H, Suarez-Lopez JR, Kayser GL. Occurrence of insecticides, fungicides, and herbicides in household drinking and irrigation water in an intensive floriculture region of Ecuador. Environmental Pollution. In Press
2. Moore RC, **Chronister BNC**, Quynh A, Yang K, Tu X, Martinez D, Suarez-Torres J, Bradman A, Suarez-Lopez JR. *Mobile vs traditional neuropsychological testing in Ecuadorian adolescents and young adults: a single- and burst-administration study in the ESPINA cohort*. Neuropsychology. 2025 Oct 23. doi: 10.1037/neu0001048. Epub ahead of print. PMID: 41129389.
3. Mehta P, Parajuli RP, **Chronister BNC**, Yang K, Tu X, Martinez D, Suarez-Torres J, Suarez JR. *Pesticide biomarkers and liver biomarkers among Ecuadorian adolescents living in agricultural settings.* Toxics. 2025;13(8):685. doi:10.3390/toxics13080685.

4. Elliott A, Parajuli RP, Mazella M, Yang K, **Chronister BNC**, Huset CA, Peterson L, Martinez D, Barr DB, Ratnani P, Gennings C, Gahagan S, Cruz F, Suarez-Torres J, Suarez-Lopez JR. *Glyphosate, dithiocarbamate*

fungicides, and PFAS in relation to body fat composition and body mass index among adolescents in agricultural communities. Environ Res. 2025;285(Pt 2):122378. doi:10.1016/j.envres.2025.122378. PMID: 40701380.

5. Parajuli RP, **Chronister BNC**, Barr DB, Suarez-Lopez JR. *Urinary pesticide biomarkers from adolescence to young adulthood in an agricultural setting in Ecuador: ESPINA study 2016 and 2022 examination data*. Data Brief. 2025;61:111882. doi:10.1016/j.dib.2025.111882. PMID: 40777557; PMCID: PMC12329275.
6. **Chronister BNC**, Kayser G, de la Cruz F, Suarez-Torres J, Lopez-Paredes D, Gahagan S, Checkoway H, Jankowska MM, Suarez-Lopez JR. *Relationships of residential distance to greenhouse floriculture and organophosphate, pyrethroid, and neonicotinoid urinary metabolite concentration in Ecuadorian adolescents.* Int J Health Geogr. 2025;24(1):9. doi:10.1186/s12942-025-00395-w.
7. Hussari M, **Chronister BNC**, Yang K, Tu X, Martinez D, Suarez-Torres J, Hong S, Suarez-Lopez JR. *Associations between pesticide metabolites and inflammatory biomarkers in adolescents and children living in an agricultural region*. Expo Health. 2025:1-18. doi:10.1007/s12403-025-00685-2.
8. Suarez-Lopez JR, Andemichael B, Yang K, **Chronister BNC,** Mackey C, Tu X, Gahagan S, Martinez D, Checkoway H, Jacobs DR, Suarez-Torres J, Hong S. *Inflammation biomarkers and neurobehavioral performance in rural adolescents*. Brain Behav Immun Health. 2025;43:100912. doi:10.1016/j.bbih.2025.100912.
9. **Chronister BNC**, Justo D, Wood RJ, Lopez-Paredes D, Gonzalez E, Suarez-Torres J, Gahagan S, Martinez D, Jacobs DR, Checkoway H, Jankowska MM, Suarez-Lopez JR. *Sex and adrenal hormones in association with insecticide biomarkers among adolescents living in Ecuadorian agricultural communities*. Int J Hyg Environ Health. 2024;259:114386. doi:10.1016/j.ijheh.2024.114386.
10. **Chronister BNC**, Yang K, Yang AR, Lin T, Tu XM, Lopez-Paredes D, Checkoway H, Suarez-Torres J, Gahagan S, Martinez D, Ospina M, Calafat AM, Barr D, Moore RC, Suarez-Lopez JR. *Glyphosate, 2,4-D, and DEET biomarkers in relation to neurobehavioral performance in Ecuadorian adolescents in the ESPINA cohort*. Environ Health Perspect. 2023;131(10):107007. doi:10.1289/EHP11383.
    a. Highlighted in: Lewis J. A thorny problem: glyphosate and 2,4-D associated with neurobehavioral effects for Ecuadorian adolescents. Environ Health Perspect. 2023;131(12):EHP14169. doi:10.1289/EHP14169.
11. Ramos SD, Kannout L, Khan H, Klasko-Foster L, **Chronister BNC**, Du Bois S. *A neighborhood-level analysis of mental health distress and income inequality as quasi-longitudinal risk of reported COVID-19 infection and mortality outcomes in Chicago*. Dialogues Health. 2023;2:100091. doi:10.1016/j.dialog.2022.100091. PMID: 36530218; PMCID: PMC9731648.
12. Forkin KT, Mitchell RD, Chiao SS, Song C, **Chronister BNC**, Wang XQ, Chisholm CA, Tiouririne M. *Impact of timing of multimodal analgesia in enhanced recovery after cesarean delivery protocols on postoperative opioids: a single center before-and-after study*. J Clin Anesth. 2022;80:110847. doi:10.1016/j.jclinane.2022.110847. PMID: 35468349.
13. **Chronister BNC**, Wu T, Santella RM, Neugut AI, Wolff MS, Chen J, Teitelbaum SL, Parada H Jr. *Dietary acid load, serum polychlorinated biphenyl levels, and mortality following breast cancer in the Long Island Breast Cancer Study Project*. Int J Environ Res Public Health. 2021;19(1):374. doi:10.3390/ijerph19010374. PMID: 35010632; PMCID: PMC8751127.
14. **Chronister BNC**, Gonzalez E, Lopez-Paredes D, Suarez-Torres J, Gahagan S, Martinez D, Barros J, Jacobs DR Jr, Checkoway H, Suarez-Lopez JR. Testosterone, estradiol, DHEA, and cortisol in relation to anxiety and depression scores in adolescents. J Affect Disord. 2021;294:838-846. doi:10.1016/j.jad.2021.07.026. PMID: 34375211; PMCID: PMC8992006.
15. Krishnan R, Deal L, Chisholm C, **Cortez B**, Boyle A. Concordance between obstetric anatomic ultrasound and fetal echocardiography in detecting congenital heart disease in high-risk pregnancies. J Ultrasound Med. 2021;40(10):2105-2112. doi:10.1002/jum.15592. PMID: 33301225.

16. Daivadanam M, Ingram M, Sidney Annerstedt K, Parker G, Bobrow K, Dolovich L, Gould G, Riddell M, Vedanthan R, Webster J, Absetz P, Mölsted Alvesson H, Androutsos O, Chavannes N, **Cortez B**, Devarasetty

P, Fottrell E, Gonzalez-Salazar F, Goudge J, Herasme O, Jennings H, Kapoor D, Kamano J, Kasteleyn MJ, Kyriakos C, Manios Y, Mogulluru K, Owolabi M, Lazo-Porras M, Silva W, Thrift A, Uvere E, Webster R, van der Kleij R, van Olmen J, Vardavas C, Zhang P. The role of context in implementation research for non-communicable diseases: answering the "how-to" dilemma. PLoS One. 2019;14(4):e0214454. doi:10.1371/journal.pone.0214454. PMID: 30958868; PMCID: PMC6453477.

Under Review

1. **Chronister BNC**, Gonzalez E, Hussari M.N, Lopez-Paredes D, Checkoway H, Suarez-Torres J, Gahagan S, Jacobs Jr. DR, Martinez D, Bradman A, Hong S, Moore RC, Suarez-Lopez JR. *Urinary Pesticide Biomarker Associations with Depression and Anxiety Scores in Adolescents and Young Adults in the ESPINA cohort.* Journal of Psychiatric Research. Review Requested.
    a. Pre-print: Chronister, Briana NC, et al. "Urinary Pesticide Biomarker Associations with Depression and Anxiety Scores from Adolescence into Young Adulthood in the ESPINA cohort." medRxiv (2025): 2025-12.
2. Kayser GL, Zhang J, Baghdadi A, Wei J, **Chronister BNC,** Calderon-Villareal A, Brouwer KB, Suarez-Torres J, Cruz F, Suarez-Lopez JR. *Sex and Adrenal Hormone Alteration in Ecuadorian Adolescents with Home and School Proximity to Floriculture Crop Area.* Environmental Health. Review Requested.
3. Hernandez R, Schneider S, Parrish E, Campbell L, Fanning J, Jin H, Roll SC, Harvey P, **Chronister BNC**, Ginsberg JP, Moore RC. *Contextual Conditions Facilitating or Compromising Momentary Cognitive Performance in Everyday Environments.* Psychological Science. Under Review.
4. Parajuli RP, Brandveen K, **Chronister BNC**, Gould CF, Martinez D, Lopez-Paredes D, Suarez-Lopez JR. *Social Support, Traditional Care Practices, and Mental Health in Rural Ecuadorian Youth*. Nature Mental Health. Under Review.
    a. Pre-print: Parajuli, Rajendra Prasad, et al. "Social Support, Traditional Care Practices, and Mental Health in Rural Ecuadorian Youth." medRxiv (2026): 2026-01.

Acknowledgements

1. Dreisbach C, Prescott S, Siega-Riz AM, McCulloch J, Habermeyer L, Dudley D, Trinchieri G, Kelsey C, Alhusen J. *Composition of the maternal gastrointestinal microbiome as a predictor of neonatal birth weight.* Pediatric Research. 2023. https://doi.org/10.1038/s41390-023-02584-4
2. Dreisbach C, Prescott S, Alhusen J, Dudley D, Trinchieri G, Siega-Riz AM. *Association between microbial composition, diversity, and function of the maternal gastrointestinal microbiome with impaired glucose tolerance on the glucose challenge test.* Plos one. 2022; 17(12), e0271261.
3. Dreisbach C, Alhusen J, Prescott S, Dudley D, Trinchieri G, Siega-Riz AM. *Metagenomic characterization of the maternal prenatal gastrointestinal microbiome by BMI*. Obesity. 2022;31(4):412-422. doi: 10.1002/oby.23659.S
4. Vedanthan R, Tuikong N, Kofler C, Blank E, Kamano JH, Naanyu V, Kimaiyo S, Inui TS, Horowitz CR, Fuster V. *Barriers, and facilitators to nurse management of hypertension: a qualitative analysis from western Kenya.* Ethnicity and Disease. 26(3), 315-322.
5. Vedanthan R, Ray M, Fuster V, Magenheim E. *Hypertension treatment rates and health care worker density: An analysis of worldwide data*. Hypertension. 2019; 73, 594–601. https://doi.org/10.1161/HYPERTENSIONAHA.118.11995

Presentations

1. Hernandez R, **Chronister BNC**, Parrish E, Harvey P, Schneider S, Campbell L, Moore RC. *Repeated and Brief Assessment of Real-World Cognitive Performance and Influencing Factors.* Poster Presentation. Accepted. International Society for CNS Clinical Trials and Methodology (ISCTM), Washington, D.C., February 2026.
2. **Chronister BNC**, Lopez-Paredes D, Checkoway H, Suarez-Torres J, Gahagan S, Martinez D, Hong S, Moore RC, Suarez-Lopez JR. *Longitudinal Associations of Urinary Pesticide Metabolites with Depression and Anxiety Symptoms from Adolescence to Young Adulthood*. Oral Presentation. Annual Conference of the International Society for Environmental Epidemiology, Atlanta, Georgia, August 2025.

3. Kayser G, Zhang J, Calderón-Villareal A, **Chronister BNC**, Suarez-Torres J, Checkoway H, De la Cruz F, Suarez-Lopez J.R. *Children's Proximity to Pesticide Spray Sites and Neurobehavioral Performance in Ecuador*. Oral Presentation. Annual Conference of the International Society for Environmental Epidemiology, Atlanta, Georgia, August 2025.
4. **Chronister BNC**, Suarez-Torres J, Martinez D, Lopez-Paredes D, Gahagan S, Sousa D, Prada Jr H, Quintana PJE, Gaines T, Moore RC, Suarez-Lopez JR. *Home proximity to agricultural crops in relation internalizing symptoms in Ecuadorian adolescents and young adults*. Poster Presentation. Annual Conference of the International Society for Environmental Epidemiology, Atlanta, Georgia, August 2025.
5. Brandveen K, Parajuli RP, **Chronister BNC**, López-Ramirez D, Suarez-Lopez J.R. *Use of social support and ancestral medicine for mental health outcomes among adolescents in the agricultural region of Pedro Moncayo, Ecuador*. Poster Presentation. Annual Conference of the International Society for Environmental Epidemiology, Atlanta, Georgia, August 2025.
6. **Chronister BNC**, Reyes S, Huezo A, Suarez-Torres J, Checkoway H, Cruz F, Moore RC, Sousa D, Suarez-Lopez JR. *Leveraging Remote Sensing and Machine Learning to Assess Pesticide Exposure from Agricultural Drift in Pedro Moncayo, Ecuador.* Poster Presentation. UC San Diego Public Health Research Day, La Jolla, California, May 5, 2025.
7. Zambrano-Romero A, Ochoa-Herrera V, Suarez-Torres J, LeFevre G, **Chronister BNC**, Suarez-Lopez JR, Kayser, G. *Pesticides and Microbial Contamination in Irrigation Systems of Pedro Moncayo-Ecuador*. Poster Presentation. UC San Diego Public Health Research Day, La Jolla, California, May 5, 2025.
8. Parajuli RP, Mehta P, **Chronister BNC**, Feldstein AE, Yang K, Suarez-Lopez JR. *Pesticide and liver biomarkers among Ecuadorians living in agrarian settings.* Poster Presentation. UC San Diego Public Health Research Day, La Jolla, California, May 5, 2025.
9. Gupta S, LeFevre G, Ochoa-Herrera V, Zambrano-Romero A, **Chronister BNC**, Mianecki A, Caprintero-Salvadore N, Checkoway H, Suarez-Torres, Suarez-Lopez JR, Kayser GL. *Neonicotinoid and atrazine pesticides in children's household drinking and irrigation water near floriculture in Pedro Moncayo, Ecuador.* Poster presentation. Consortium of Universities for Globa Health, Atlanta, GA, February 20-23, 2025.
10. **Chronister BNC**, Reyes S, Huezo A, Suarez-Torres J, Checkoway H, Cruz F, Moore R.C., Sousa D, Suarez-Lopez JR. *Leveraging Remote Sensing And Uniform Manifold Approximation And Projection To Assess The Impact Of Home Proximity to Agriculture on Pesticide Exposure in an Ecuadorian Agricultural Community*. Poster Presentation. American Geophysical Union (AGU) Conference, Washington, D.C., December 9-13, 2024.
11. **Chronister BNC**, Reyes S, Sousa D. *Using Manifold Learning and Segment Anything to identify small- and large-scale agriculture in an Equatorial Agricultural Community*. Poster Presentation. American Geophysical Union (AGU) Conference, Washington, D.C., December 9-13, 2024.
12. **Chronister BNC**, Quynh A, Yang K, Tu X, Martinez D, Suarez-Torres J, Bradman A, Suarez-Lopez JR, Moore RC. *Comparing Mobile Cognitive Testing with Traditional Neuropsychological Testing in Ecuadorian Adolescent and Young Adults in the ESPINA Cohort: A Single- and Burst-Administration Study.* Poster Presentation. Technology in Psychiatry Summit (TIPS), Phoenix, Arizona, December 7th, 2024.
13. **Chronister BNC**, Kayser G, de la Cruz F, Suarez-Torres J, Lopez-Paredes D, Gahagan S, Checkoway H, Jankowska MM, Suarez-Lopez JR. *Home distance to Greenhouse Floriculture and Organophosphate, Pyrethroid, and Neonicotinoid Urinary Metabolite Concentration in Ecuadorian Adolescents*. Oral Presentation. Annual Conference of the International Society for Environmental Epidemiology, Santiago, Chile, August 2024.
14. Hussari MN, **Chronister BNC**, Yang K, Tu X, Parajuli R, Martinez D, Suarez-Torres J, Hong S, Suarez-Lopez JR. Associations between Pesticide Metabolites and Inflammatory Biomarkers in Adolescents and Children Living in an Agricultural Region. Oral Presentation. Annual Conference of the International Society for Environmental Epidemiology, Santiago, Chile, August 2024.
15. **Chronister BNC**, Kayser G, de la Cruz F, Suarez-Torres J, Lopez-Paredes D, Gahagan S, Checkoway H, Jankowska MM, Suarez-Lopez JR. *Home distance to Greenhouse Floricultural and Organophosphate, Pyrethroid, and Neonicotinoid Urinary Metabolite Concentration in Ecuadorian Adolescents*. Poster Presentation. University of California San Diego Public Health Research Day, May 2024.
16. Hussari M, **Chronister BNC**, Yang K, Tu X, Martinez D, Suarez-Torres J, Hong S, Suarez-Lopez JR. *Associations between Pesticide Metabolites and Inflammatory Biomarkers in Adolescents and Children*

Living in an Agricultural Region. Poster Presentation. UC San Diego Public Health Research Day, May 2024. Received Outstanding Poster Award.

17. **Chronister BNC**, Kayser G, de la Cruz F, Suarez-Torres J, Lopez-Paredes D, Gahagan S, Checkoway H, Jankowska M, Suarez-Lopez JR. *Home distance to Greenhouse Floricultural and Organophosphate, Pyrethroid, and Neonicotinoid Urinary Metabolite Concentration in Ecuadorian Adolescents*. Oral Presentation. UCSD Integrated Cardiovascular Epidemiology Fellowship and the Advanced Data Analytics Training for Behavioral and Social Sciences Research (TADA-BSSR) Joint Research Symposium, October 2023
18. **Chronister BNC**, Gonzalez E, Contreras A, Lopez-Paredes D, Checkoway H, Suarez-Torres J, Gahagan S, Jacobs Jr. DR, Martinez D, Suarez-Lopez JR. *Associations of Pesticide Exposure with Depression and Anxiety in Ecuadorian Adolescents, and Effect Modification by Social Determinants of Health*. Poster Presentation. Annual Conference of the International Society for Environmental Epidemiology, September 2023.
19. **Chronister BNC**, Justo D, Lopez-Paredes D, Gonzalez E, Suarez-Torres J, Gahagan S, Martinez D, Jacobs DR, Checkoway H, Suarez-Lopez JR. *Urinary Organophosphate and Neonicotinoid Metabolites in relation to Sex and Adrenal Hormones among Adolescents living in Ecuador*. Accepted. Oral Presentation. Annual Conference of the International Society for Environmental Epidemiology, September 2023.
20. **Chronister BNC**, Jankowska MM, Kayser G, Suarez-Torres J, Gahagan S, Checkoway H, Suarez-Lopez JR. *Geospatial determinants of urinary insecticide concentrations and internalizing disorders in Ecuadorian adolescents*. Presented. Oral Presentation. Annual Conference of the International Society for Environmental Epidemiology, October 2022.
21. Kayser G, Zhang J, Wei J, **Chronister BNC**, Suarez-Lopez JR. *Residential Proximity to Pesticide Sprayed Sites and Sex and Adrenal Hormones in Ecuadorian Adolescents.* Presented. Oral Presentation. Annual Conference of the International Society for Environmental Epidemiology, October 2022.
22. Chairs: **Chronister BNC**, Suarez-Lopez JR; Presenters: Ritz B, Kayser G, **Chronister BNC**, Simões M, Gunier R, Palomo L. *Location, Location, Location! The importance of geospatial exposure constructs to assess pesticide exposures and related adverse health outcomes in agricultural settings.* Facilitated. Symposium. International Society for Environmental Epidemiology Conference, October 2022.
23. **Chronister BNC**, Justo D, Lopez-Paredes D, Gonzalez E, Suarez-Torres J, Gahagan S, Martinez D, Jacobs Jr. DR, Checkoway H, Suarez-Lopez JR. *Urinary Organophosphate and Neonicotinoid Metabolites in relation to Sex and Adrenal Hormones among Adolescents living in Ecuadorian Agricultural Communities*. Poster Presentation. Society for Epidemiologic Research Conference, Chicago, IL, USA. June 14-17, 2022.
24. **Chronister BNC**, Gonzalez E, Contreras A, Lopez-Paredes D, Checkoway H, Suarez-Torres J, Gahagan S, Jacobs Jr. DR, Martinez D, Suarez-Lopez JR. (2022, April 21-22). *Associations of Pesticide Exposure with Depression and Anxiety in Ecuadorian Adolescents, and Effect Modification by Social Determinants of Health*. Accepted. Oral Presentation. Socio-Environmental Research Group (SERG) Conference, April 2022.
25. Suarez JR, **Chronister BNC,** Kayser G, Suarez-Torres J, Checkoway H, Friedman E, Martinez D. *Residential proximity to crops in relation to pesticide exposure and mental health in Ecuadorian children and adolescents: The ESPINA study*. Presented. Oral Presentation. International Congress on Occupational Health, December 2021.
26. **Chronister BNC,** Justo D, Lopez-Paredes D, Gonzalez G, Suarez-Torres J, Gahagan S, Martinez D, Jacobs Jr DR, Checkoway H, Suarez-Lopez JR. *Associations of Glyphosate with Testosterone, Cortisol, DHEA, and Estradiol in Ecuadorian Adolescents.* Presented. Oral Presentation. International Society of Environmental Exposures Conference, August 2021.
27. Suarez-Lopez JR, **Chronister BNC**, Yang K., Tu X, Lopez-Paredes D, Checkoway H, Suarez-Torres J, Gahagan S, Jacobs Jr. DR. *Associations of herbicides and DEET repellent metabolites with neurobehavioral performance in adolescents.* Presented. Oral Presentation. International Society of Environmental Exposures Conference, August 2021.
28. Gonzalez E, **Chronister BNC**, Sadler GR, Suarez-Lopez JR. *Gastric and lymphatic-hematopoietic tissue cancer mortality across provinces in Ecuador, 2010 – 2020.* Oral presentation. UC San Diego Summer Research Conference. Virtual, August 12, 2021.
29. **Chronister BNC**, Wu T, Santella RM, Neugut AI, Wolff MS, Teitelbaum SL, Parada Jr H. *Dietary acid load, serum polychlorinated biphenyl (PCB) levels, and mortality following breast cancer in the Long Island Breast Cancer Study Project*. Presented. Poster Presentation. March 2021. American Society of Preventative

Oncology Conference, Virtual, March 2021.

30. Gonzalez E, **Chronister B**, Nguyen-Grozavu F, Suarez-Lopez J. *Pesticide Exposure: A Risk Factor for Childhood Leukemia among Hispanic American.* Oral Presentation. International Cancer Education Conference, Virtual Conference, October 2020.
31. **Chronister B**, Lopez-Paredes D, Gonzalez E, Suarez-Torres J, Gahagan S, Martinez D, Barros J, Checkoway H, Suarez-Lopez J. *The Effects of Testosterone, Estradiol, Dehydroepiandrosterone, and Cortisol on Anxiety and Depression Symptoms in Ecuadorian Adolescents.* Poster Presentation. American Academy of Pediatrics Conference, Virtual Conference, October 2020.
32. Gonzalez E, **Chronister B**, Nguyen-Grozavu F, Binda A, Suarez-Lopez JR, Sadler GR. *Linking pesticide exposure to childhood leukemia among Hispanic Americans*. Oral Presentation. UC San Diego Summer Research Conference, virtual, August 13, 2020.
33. Gammon B, Melanson N, **Chronister B**, Urban A, Chisholm C, Ferguson J, Dudley D. *Validation of a cesarean delivery risk calculator in women undergoing induction of labor.* Poster Presentation. American Congress of Obstetricians and Gynecologists Conference, Seattle, Washington, April 2020.
34. Whitham M, Gammon E, **Chronister B**, Urban A, Silver R, Reddy U, Dudley D. Risk factors for recurrent stillbirth (SB): results from the Stillbirth Collaborative Research Network (SCRN). Poster Presentation. Society for Maternal and Fetal Medicine Conference, Grapevine, Texas, February 2020.
35. Mitchell R, **Chronister BN**, Chisholm CA, Tiouririne M. *Pain management and opioid utilization after updating the enhanced recovery after surgery for scheduled cesarean*. Poster Presentation. The Anesthesiology Annual Meeting, Orlando, Florida, October 2019.
36. Podwika S, Wang C, Zuber MJ, Chisholm C, **Cortez B**, Dudley D. *Morbidly Adherent Placenta Interdisciplinary Management*. Poster Presentation. American Congress of Obstetricians and Gynecologists Conference, Orlando, Florida, May 2019
37. Zuber MJ, Dudley D, Carlson S, Cotton-Caballero A, **Cortez B**. *Urine protein/creatinine ratio (UPCR) at term: a prospective cohort study*. Poster Presentation. American Congress of Obstetricians and Gynecologists Conference, Orlando, Florida, May 2019
38. Lane MA, Dudley D, **Cortez B**, Boyle A, Ferguson J. *Antenatal Predictors of NICU Admission at a High-Risk Perinatal Center*. Poster Presentation. American Congress of Obstetricians and Gynecologists Conference, Orlando, Florida, May 2019
39. Scrivani C, **Cortez B**, Chisholm C, Boyle A, Ferguson J, Kirk S, Dudley D. *Maternal Complications in Type 2 Diabetes Mellitus Treated with Combination Compared to Single Agent Therapy*. Poster Presentation. American Congress of Obstetricians and Gynecologists Conference, Orlando, Florida, May 2019
40. Krishnan R, Deal L, **Cortez B**, Mitchell R, Chisholm C, Boyle A. *Concordance Between Obstetric Anatomic Ultrasound and Fetal Echocardiography in Diagnosing Fetal Cardiac Anomalies*. Oral Presentation. American Institute of Ultrasound in Medicine Convention, Orlando, Florida, April 2019.
41. Deal L, Dudley D, **Cortez B**, Silver R, Varner M, Saade G, Stoll B, Bukowski R, Conway D, Hogue C. *Stillbirth and Prenatal Vitamin Supplementation: Analysis from the Stillbirth Collaborative Research Network (SCRN).* Poster Presentation. Society for Maternal and Fetal Medicine Conference, Las Vegas, Nevada, February 2019.
42. Fernandez-Jimenez R, Latina J, Soto AV, Bansilal S, Vedanthan R, Jaslow R, Santana M, Diaz-Munoz R, Regazzoni V, **Cortez B**, Clarke-Littman A, Hernandez E, Fatterpekar M, Shulman A, Giannarelli C, Kovacic J, Kasarskis A, Fayad Z, Hajjar R, Fuster V. *Social Determinants of lifestyle and Cardiovascular Health in a Minority Community Study for Promotion of Health*. Poster Presentation. American College of Cardiology (ACC) Annual Scientific Sessions. March 2018.
43. Zuber MJ, **Cortez B**, Chisholm C. *Assessment of Cystoscopy Rates During Cesarean Delivery at a High-risk Perinatal Center.* Poster Presentation. American Congress of Obstetricians and Gynecologists District IV Meeting, Savannah, Georgia, September 2018.
44. Latina JM, Fernandez-Jiminez R, Bansilal S, Vedanthan R, Soto A, Jaslow R, Santana M, **Cortez B**, Schulman A, Fatterpekar M, Hajjar R, Kasarskis A, Fayad ZA, Fuster V. *Ethnic disparities in metabolic syndrome among caregivers of preschool children in a community-based trial.* Poster Presentation. American Heart Association Scientific Sessions, Anaheim, California, November 2017.
45. **Cortez B**, Campbell K, Sigala A, Beidinger H, Haussaman L, Taylor K, Lee S. *Implementation of an antibiotic education program for community health promoters and an assessment of antibiotic knowledge in Tena and Quito, Ecuador*. Poster Presentation, Global Health Forum, Notre Dame, Indiana, July 2015.

PENDING GRANT SUPPORT

2026-2031 **NIEHS K99/R00**
Linking Pesticide Exposure to Depression, Anxiety, and Cognitive Function Using Mobility Based Geospatial Models and Multi-Omics Approaches
Role: Principal Investigator

SERVICE FOR THE UNIVERSITY (University of California San Diego)

AY 2025 – 2026 Public Health Research Day Planning Committee Member

WORKSHOPS ATTENDED

1. Leveraging Geospatial Technologies for Environmental Health Decisions (April 14-15, 2021)
2. The Interplay between Environmental Exposures and Mental Health Outcomes (February 2-3, 2021)
3. Columbia Environmental Mixtures Workshop (July 28th-29th, 2022)
4. The Institute of Electrical and Electronics Engineers (IEEE) Geoscience and Remote Sensing Society (GRSS)-University of Southern California Ming Hsieh Institute Remote Sensing Summer School (July 13-15, 2023)

CERTIFICATIONS

1. Certificate in Spatial Analysis, University of California San Diego, Spring 2021.
   - Coursework included:
     1. GPEC 443: GIS and Spatial Data Analysis
     2. GPEC 444: Advanced GIS and Remote Sensing
     3. GPGN 490: Geospatial Analysis with R
2. Certificate of Completion: IGARSS 2023 Summer School: Introduction to Geospatial Raster and Vector Data with Python
3. Certificate of Completion: IGARSS 2023 Summer School: Deep Learning for Remote Sensing Data Analysis, July 14, 2023
4. Certificate of Completion: IGARSS 2023 Summer School: Introduction to Radar Interferometry and its Applications, July 15, 2023
5. Certificate in Machine Learning, UC San Diego Extended Studies, In Progress.

MENTORING

2025-Present An Nguyen, UCSD, Bachelors in Public Health Student
2025-Present Naimisha Adira, UCSD, Master's in Public Health Student
2021-2022 Priyanka Mehta, UCSD, Master's Student Capstone Committee Member
2020-2023 Andrea Cardenas, UCSD, Bachelors in Public Health Research Project
2019-2021 Eduardo Gonzalez, UCSD, Bachelors in Public Health Research Project

INVITED LECTURES

San Marcos State University: Leveraging Remote Sensing and Machine Learning to Assess the Impact of Home Proximity to Agriculture on Pesticide Exposure, Depression, and Anxiety in an Ecuadorian Agricultural Community, February 10th, 2025
San Diego State University: Introduction to Macros in SAS, March 15, 2021, November 3, 2022, November 20, 2023, and November 15, 2024, November 17, 2025
San Diego State University: Implementing an Antibiotic Resistance Education Program in Ecuador, March 17, 2022

HONORS AND AWARDS

Recipient, *Travel Award*, SERG Conference, 2022

Trainee, *OBSSR T32 Training in Advanced Data Analytics for Behavioral and Social Sciences Grant,* UCSD, 2021-2024

Scholar, *Hispanic Center of Excellence*, San Diego State University, 2021-2022

Recipient, *Tinker Field Research Travel Grant,* University of California San Diego, 2020

Recipient, *Global Health Student Field Experience/Research Award,* University of California San Diego, 2020

Scholar, *Hispanic Center of Excellence*, San Diego State University, 2020-2021

Recipient, *Global Health Travel Award, University of Notre Dame*, 2015

Recipient, *Global Health Scholarship*, University of Notre Dame, 2014

Recipient, *Dooley Society Medical Mission Stipend Award*, University of Notre Dame, 2013

Recipient, *James F. Andrews Scholarship*, University of Notre Dame, 2013

TECHNICAL SKILLS

COMPUTER SKILLS:

- Advanced: R, SAS, QGIS, ArcGIS Pro, REDCap
- Intermediate: Python, STATA, Google Earth Engine, NVivo, ENVI

SKILLS: Qualitative data collection and analysis (e.g., focus groups, transcription), quantitative data collection and analysis, survey design, development and implementation of health interventions, research project management, participant recruitment and enrollment, basic phlebotomy.

LAB SKILLS: Fluorescent microscopy, light microscopy aseptic technique, spectrophotometer, infrared spectroscopy, southern and western blotting, gram staining, mammalian cell culture, bacteria cultures, colony isolation, PCR, DNA extraction, DNA cloning, gel electrophoresis, phlebotomy, MCF-7, and MCF-10 work.

LANGUAGES: Fluent in Spanish